



AQ9.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

Functions of several variables:

Level 1:

1 point

Which of the following are correct with respect to $f(x, y) = x^2 - 2y$, where x, y are real?

☐

Domain of f is $\mathbb{R}^2$

☐

Domain of f is $\mathbb{R}$

☐

Range of f is $\mathbb{R}$

☐

Range of f is $\mathbb{R}^2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Domain of f is $\mathbb{R}^2$

Range of f is $\mathbb{R}$

1 point

Let $D \subset \mathbb{R}^n$ and $f: D \rightarrow \mathbb{R}^m$, $g: D \rightarrow \mathbb{R}^m$ be multivariable functions on D . Which of the following statements is(are) TRUE?

☐

The product function $f \cdot g$ is defined on D by $(f \cdot g)(x) = f(x) \times g(x)$, $x \in D$

☐

The sum function $f + g$ is defined on D by $(f + g)(x) = f(x) + g(x)$, $x \in D$

☐

Let $c \in \mathbb{R}$. The function cf is defined on D by $(cf)(x) = c \times f(x)$, $x \in D$

☐

If $m = 1$ and $g(x) \neq 0$, $x \in D$, then the function f/g is defined on D by $(f/g)(x) = f(x)/g(x)$, $x \in D$

No, the answer is incorrect.

Score: 0

Accepted Answers:

The sum function $f + g$ is defined on D by $(f + g)(x) = f(x) + g(x)$, $x \in D$



If $m = 1$ and $g(x) \neq 0$, $x \in D$, then the function f/g is defined on D by $(f/g)(x) = f(x)/g(x)$, $x \in D$.

1 point

Suppose $f: D \rightarrow \mathbb{R}$ is a function defined on a domain $D \subset \mathbb{R}$. Which of the following can be f ?

☐

$$f(x) = 4 \pm \sqrt{x-10} \quad f(x) = 4 \pm x - 10$$

☐

$$f(x) = \frac{x^2}{x+1} \quad f(x) = x+1x^2$$

☐

$$f(x) = \log(\sqrt{x^4 + 3}) \quad f(x) = \log(x^4 + 3)$$

☐

$$f(x) = x^3 + x^2 - x \quad f(x) = x^3 + x^2 - x$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x) = \frac{x^2}{x+1} \quad f(x) = x+1x^2$$

$$f(x) = \log(\sqrt{x^4 + 3}) \quad f(x) = \log(x^4 + 3)$$

☐

$$f(x) = x^3 + x^2 - x \quad f(x) = x^3 + x^2 - x$$

1 point

Which of the following statements is(are) TRUE?

☐

The output of a scalar valued multivariable function is always a real number.

☐

The output of a vector valued multivariable function can be a real number.

☐

The output of a single variable vector valued function is always a vector.

☐

The output of a vector valued multivariable function can not be a real number.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The output of a scalar valued multivariable function is always a real number.

The output of a single variable vector valued function is always a vector.

The output of a vector valued multivariable function can not be a real number.

1 point

Consider the function $S(t) = (\cos t, \sin t, 2t)$, where t is real. Choose the set of correct options.

☐

S is a single-variable vector-valued function

☐

S is a multi-variable vector-valued function

☐

The domain of S is $\mathbb{R}$ and the co-domain of S is $\mathbb{R}^3$

☐

The domain of S is $\mathbb{R}^3$ and the co-domain of S is $\mathbb{R}^3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

SS is a single-variable vector-valued function

The domain of SS is $\mathbb{R}$ and the co-domain of SS is $\mathbb{R}^3$

How many of the following functions is(are) a vector valued multivariable function?

- $f(x, y) = \frac{1}{3\pi} e^{x^2 + y^2}$ • $f(x, y) = 3\pi 1 e^{x^2 + y^2}$
- $f(x, y) = (x^3, y^2)$ • $f(x, y) = (x^3, y^2)$
- $f(x, y) = (x + y, x - y, 2xy)$ • $f(x, y) = (x + y, x - y, 2xy)$
- $f(x, y, z) = x^2 + xyz - z^2$ • $f(x, y, z) = x^2 + xyz - z^2$
- $f(x) = (x + 1, x - 1)$ • $f(x) = (x + 1, x - 1)$
- $f(x, y, z) = (e^{x+y+z}, 0, 0)$ • $f(x, y, z) = (e^{x+y+z}, 0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

1 point

Suppose $g: D \rightarrow \mathbb{R}^3$ $g: D \rightarrow \mathbb{R}^3$ is a function defined on a domain $D \subset \mathbb{R}^2$ $D \subset \mathbb{R}^2$. Which of the following functions may be gg ?

☐

$$g(x, y) = 4x + 3y - 12xy \quad g(x, y) = 4x + 3y - 12xy$$

☐

$$g(x, y) = (x^4 + 1, y^4 + 1, xy) \quad g(x, y) = (x^4 + 1, y^4 + 1, xy)$$

☐

$$g(x, y, z) = (e^x - e^z, e^z - e^y) \quad g(x, y, z) = (e^x - e^z, e^z - e^y)$$

☐

$$g(x, y) = (\sin(x + y), \cos(x - y)) \quad g(x, y) = (\sin(x + y), \cos(x - y))$$

☐

$$g(x, y) = (0, \sin(e^x), \cos(e^x)) \quad g(x, y) = (0, \sin(e^x), \cos(e^x))$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$g(x, y) = (x^4 + 1, y^4 + 1, xy) \quad g(x, y) = (x^4 + 1, y^4 + 1, xy)$$

$$g(x, y) = (0, \sin(e^x), \cos(e^x)) \quad g(x, y) = (0, \sin(e^x), \cos(e^x))$$

Level 2:

1 point

Which of the following are correct with respect to $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{16}}$ $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{16}}$

☐

, where x, y are real?

☐

Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$ $D = \{(x, y) \mid 9x^2 + 16y^2 \leq 1\}$

☐

Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \geq 1\}$ $D = \{(x, y) \mid 9x^2 + 16y^2 \geq 1\}$

☐

Range of f is $[0, 1][0, 1]$

☐

Range of f is $[0, 1)[0, 1)$

☐

Range of f is $[1, \infty)[1, \infty)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$ $D = \{(x, y) \mid 9x^2 + 16y^2 \leq 1\}$

Range of f is $[0, 1][0, 1]$

1 point

A function $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ has the form $F(x, y) = (P(x, y), Q(x, y), R(x, y))$ $F(x, y) = (P(x, y), Q(x, y), R(x, y))$. Choose the correct option(s).

☐

F is a multi-variable vector-valued function

☐

F is a single-variable vector-valued function

☐

P, Q, R, P, Q, R are single-variable vector-valued functions

☐

P, Q, R, P, Q, R are multi-variable vector-valued functions

☐

P, Q, R, P, Q, R are multi-variable scalar-valued functions

No, the answer is incorrect.

Score: 0

Accepted Answers:

F is a multi-variable vector-valued function

P, Q, R, P, Q, R are multi-variable scalar-valued functions

1 point

Consider $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{4}}$ $f(x, y) = \sqrt{1 - 9x^2 - 4y^2}$

☐

. Note: An ellipse is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $a^2x^2 + b^2y^2 = 1$ where a, b are the dimensions of the ellipse.

☐

Ellipses of increasing size represent the curves $\{(x, y) \mid f(x, y) = c\}$ $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1

☐

Ellipses of decreasing size represent the curves $\{(x, y) \mid f(x, y) = c\}$ $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1

☐

Ellipses of constant size represent the curves $\{(x, y) \mid f(x, y) = c\}$ $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1

☐

The curves $\{(x, y) \mid f(x, y) = c\}$ $\{(x, y) \mid f(x, y) = c\}$, where $c \in [0, 1)$ $c \in [0, 1)$, cannot be represented by an ellipse

☐

$f(x, y) = 1$ is a point

No, the answer is incorrect.

Score: 0

Accepted Answers:

Ellipses of decreasing size represent the curves $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1

$f(x, y) = 1$ is a point

1 point

Consider a point source $S = (1, 2, 3)$ radiating energy. The intensity I at a given point

$P = (x, y, z)$ in space is inversely related to the square of the distance d between S and

P : $I(x, y, z) = \frac{k}{d^2}$, where k is a real positive constant. Choose the correct option(s).

☐

Intensity I is constant on $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 1\}$

☐

Intensity I is constant on $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 2\}$

☐

Intensity I is constant on $\{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$

☐

Intensity I is constant on $\{(x, y, z) \mid x^2 + y^2 + z^2 = 2\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Intensity I is constant on $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 1\}$

Intensity I is constant on $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 2\}$